

The Constrained Optimisation Problem

EC306: Intermediate Microeconomics — Day 2

Core reading: Perloff (2020), Ch. 4

Recap and today's move

- **Yesterday:** what the consumer *wants* — preferences, indifference curves, the MRS, and utility as an ordinal summary.
- **Today:** what the consumer *can afford*, and how wants and means collide to pin down a single chosen bundle.

The whole problem in one line

$$\max_{x,y} u(x,y) \quad \text{subject to} \quad p_x x + p_y y \leq m.$$

Everything today is setting up this problem, solving it, and reading the economics off the solution.

Roadmap

- ➊ The budget set and budget line
- ➋ The consumer's problem: objective and constraint
- ➌ The Lagrangian — set up, solve, read the optimality condition
- ➍ “Bang per buck” and the meaning of λ
- ➎ Interior solution: Cobb–Douglas demand
- ➏ Special cases: perfect substitutes, perfect complements, quasilinear
- ➐ Graphical summary of every case

The budget set: what money lets you buy

With income m and prices p_x, p_y , a bundle (x, y) is **affordable** when spending does not exceed income:

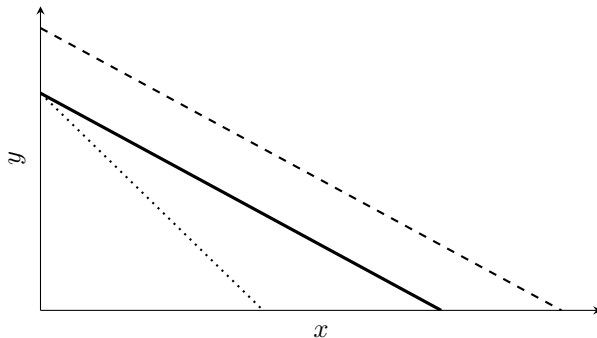
$$\underbrace{p_x x + p_y y}_{\text{spending}} \leq \underbrace{m}_{\text{income}} .$$

The set of all such bundles is the **budget set**. Because more is better (monotonicity), the consumer spends *everything* and lands on the boundary — the **budget line**:

$$p_x x + p_y y = m \quad \Longleftrightarrow \quad y = \frac{m}{p_y} - \frac{p_x}{p_y} x .$$

- **Intercepts:** $\frac{m}{p_x}$ (all income on x) and $\frac{m}{p_y}$ (all income on y).
- **Slope** $= -\frac{p_x}{p_y}$: the *market* rate of exchange — to get one more x you must give up p_x/p_y units of y (its opportunity cost). Hold this next to the MRS, the rate you are *willing* to trade. The optimum is where the two agree.

What moves the budget line



Solid — baseline. **Dashed** — income $m \uparrow$: parallel *outward shift*, slope unchanged. **Dotted** — $p_x \uparrow$: *pivots inward* about the y -intercept (x alone got dearer, so m/p_x falls).

Why pivots matter

A pure price change *rotates* the line: relative prices, not just purchasing power, have changed. This is the seed of the substitution effect we unpack tomorrow.

The consumer's problem

The consumer chooses (x, y) to climb as high as possible on the preference map without overspending:

$$\max_{x, y} \underbrace{u(x, y)}_{\text{objective function}} \quad \text{subject to} \quad \underbrace{p_x x + p_y y \leq m}_{\text{constraint}}.$$

- The **objective function** $u(x, y)$ is what we push *up* — it ranks bundles by desirability.
- The **constraint** $p_x x + p_y y \leq m$ is what holds us *back* — the affordable set.
- **Choice variables:** x, y . **Parameters** (fixed for the consumer): p_x, p_y, m .

Why this needs a special tool

We maximise one function while a *second* equation ties the variables together. Plain “set the derivative to zero” fails — nudging x forces y to move through the budget. The **Lagrangian** builds that link directly into the objective.

Step 1 — Set up the Lagrangian

Write the constraint in slack form, $m - p_x x - p_y y = 0$, and attach it to the objective with a multiplier λ :

$$\mathcal{L}(x, y, \lambda) = \underbrace{u(x, y)}_{\text{objective}} + \lambda \underbrace{(m - p_x x - p_y y)}_{\text{constraint} = 0}.$$

What λ is doing

λ is the “price” the problem pays for breaking the budget. When the constraint binds, the bracket is 0, so the added term contributes nothing to $\mathcal{L}$ at the optimum. Maximising $\mathcal{L}$ freely over the *three* variables (x, y, λ) is therefore equivalent to maximising u *along* the budget line — but now without an explicit constraint.

Step 2 — Solve: the first-order conditions

Set each partial derivative of $\mathcal{L}$ to zero, writing $MU_x = \partial u / \partial x$ and $MU_y = \partial u / \partial y$:

$$\frac{\partial \mathcal{L}}{\partial x} = MU_x - \lambda p_x = 0 \implies MU_x = \lambda p_x,$$

$$\frac{\partial \mathcal{L}}{\partial y} = MU_y - \lambda p_y = 0 \implies MU_y = \lambda p_y,$$

$$\frac{\partial \mathcal{L}}{\partial \lambda} = m - p_x x - p_y y = 0 \implies p_x x + p_y y = m.$$

Three equations, three unknowns (x, y, λ) . The third simply returns the budget, holding with equality — exactly as monotonicity predicted.

Step 3 — The optimality condition

Divide the first FOC by the second to eliminate λ :

$$\frac{MU_x}{MU_y} = \frac{\lambda p_x}{\lambda p_y} = \frac{p_x}{p_y}.$$

The left-hand side is the **MRS**, so at an interior optimum:

$$\text{MRS} = \frac{MU_x}{MU_y} = \frac{p_x}{p_y}$$

Tangency, in words

Willing trade-off (MRS, slope of the indifference curve) = *market* trade-off (p_x/p_y , slope of the budget line). The indifference curve is **tangent** to the budget line. Pair this equation with the budget to pin down (x^*, y^*) .

“Bang per buck” and the meaning of λ

Now rearrange the *same* two FOCs the other way — solve each for λ :

$$\lambda = \frac{\text{MU}_x}{p_x} = \frac{\text{MU}_y}{p_y}.$$

Equal marginal utility per pound

MU_x/p_x is the extra utility from the *last pound* spent on x — its “bang per buck”. At the optimum every good delivers the *same* bang per buck. If x paid more, you would shift a pound from y to x and gain; at the optimum no such gain remains.

So what is λ ?

That common value *is* λ : the **marginal utility of income**, $\lambda = \partial u^* / \partial m$ — the utility bought by one more pound. The same object reappears as the **shadow price** of a binding constraint throughout economics.

Worked example — Cobb–Douglas demand

Let $u = x^a y^b$ with budget $p_x x + p_y y = m$. **Step 3 (tangency):**

$$\frac{MU_x}{MU_y} = \frac{ax^{a-1}y^b}{bx^ay^{b-1}} = \frac{ay}{bx} = \frac{p_x}{p_y} \implies p_y y = \frac{b}{a} p_x x.$$

Substitute into the budget $p_x x + \frac{b}{a} p_x x = m$:

$$x^* = \frac{a}{a+b} \frac{m}{p_x}, \quad y^* = \frac{b}{a+b} \frac{m}{p_y}.$$

The expenditure-share result

Spending on x is $p_x x^* = \frac{a}{a+b} m$ — a *constant fraction* of income, independent of prices. Stable budget shares are why Cobb–Douglas is the empirical workhorse.

Your turn (1)

Your turn

$u = x^{0.25}y^{0.75}$, $m = 120$, $p_x = 2$, $p_y = 3$. Find x^* , y^* , and the share of income spent on each good.

Perfect substitutes — a constant trade-off

$u = ax + by$. The $MRS = a/b$ is *constant*: indifference curves are straight lines, so tangency generically cannot hold. Compare bang per buck directly:

$$\frac{MU_x}{p_x} = \frac{a}{p_x} \quad \text{vs.} \quad \frac{MU_y}{p_y} = \frac{b}{p_y}.$$

Three cases:

- $\frac{a}{p_x} > \frac{b}{p_y}$ (x is the better deal everywhere) $\Rightarrow$ **corner** $x^* = \frac{m}{p_x}$, $y^* = 0$.
- $\frac{a}{p_x} < \frac{b}{p_y}$ (y is the better deal everywhere) $\Rightarrow$ **corner** $x^* = 0$, $y^* = \frac{m}{p_y}$.
- $\frac{a}{p_x} = \frac{b}{p_y}$ (equally good) $\Rightarrow$ **any** bundle on the budget line is optimal.

Quick example

$u = 2x + y$, $m = 4$, $p_x = p_y = 1$: bang per buck $\frac{a}{p_x} = 2 > \frac{b}{p_y} = 1$, so spend all on x : $x^* = 4$, $y^* = 0$.

Perfect complements — fixed proportions

$u = \min\{x, y\}$ (left and right shoes): the goods are used together, so extra of one alone adds *nothing*. Indifference curves are L-shaped, with kinks on the ray $y = x$. There is no interior tangency — the optimum sits at a **kink**.

Step 1 — proportion condition. Overshooting one good is pure waste, so the optimum uses equal amounts:

$$x = y.$$

Step 2 — impose the budget $p_x x + p_y y = m$ with $y = x$:

$$(p_x + p_y) x = m \implies \boxed{x^* = y^* = \frac{m}{p_x + p_y}.$$

Quick example

$$u = \min\{x, y\}, m = 60, p_x = 2, p_y = 4: \quad x^* = y^* = \frac{60}{2 + 4} = 10.$$

Your turn (2) — quasilinear utility

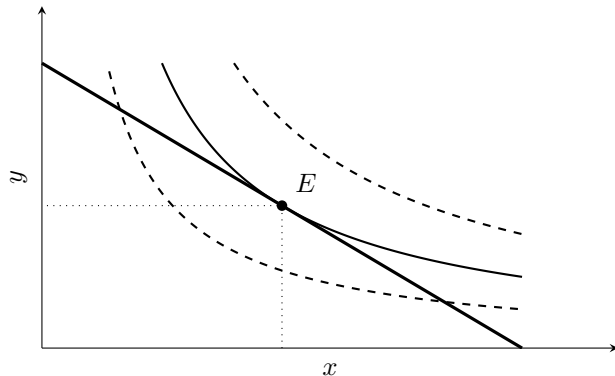
A **quasilinear** utility is linear in one good: $u = v(x) + y$ with v concave. The y -term has constant marginal utility, so x^* is set by *prices alone* (no income effect on x) and spare income flows entirely to y .

Your turn

$$u = 4\sqrt{x} + y, \quad m = 8, \quad p_x = 1, \quad p_y = 1.$$

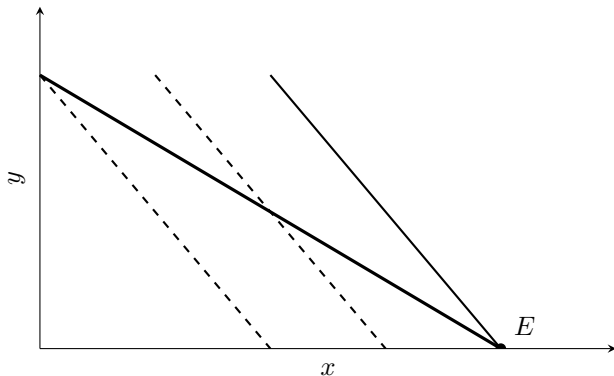
- 1 Use $MU_x/MU_y = p_x/p_y$ to find x^* .
- 2 Use the budget to find y^* .
- 3 Check $y^* \geq 0$ (otherwise you are at the corner $y = 0$, $x^* = m/p_x$).

Graph 1 — Interior solution (tangency)



Straight line: budget. Solid curve: the highest reachable IC, **tangent** at E (single touch). Inner dashed: affordable but lower utility. Outer dashed: preferred but unaffordable.

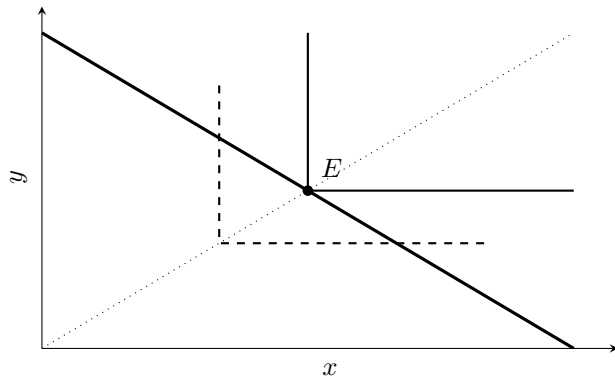
Graph 2 — Perfect substitutes (corner)



Indifference curves (slope $-a/b = -2$) are *steeper* than the budget (slope -1), so x pays more per pound everywhere.

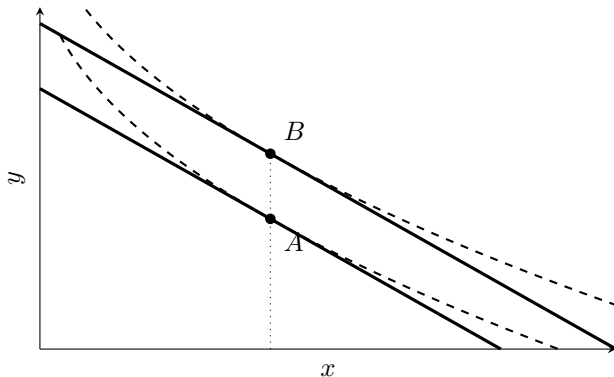
The best IC reaches the feasible set only at the x -intercept: corner $E = (m/p_x, 0) = (4, 0)$.

Graph 3 — Perfect complements (kink)



L-shaped ICs kink on the ray $y = x$ (dotted). The optimum is where the budget meets that ray: $E = (3, 3)$, i.e. $x^* = y^* = m/(p_x + p_y)$. Solid L: best affordable. Dashed L: lower utility.

Graph 4 — Quasilinear (no income effect on x)



Income rises ($m : 8 \rightarrow 10$): the budget shifts out and the optimum moves $A \rightarrow B$ straight *up*. Both tangencies sit at $x^* = 4$ — all extra income goes to y , none to x .

Summary — one method, four faces

The method

Set up $\mathcal{L} = u + \lambda(m - p_x x - p_y y)$. The FOCs give either the tangency form $MRS = p_x/p_y$ or the bang-per-buck form $MU_x/p_x = MU_y/p_y = \lambda$. Pair with the budget to solve; λ is the marginal utility of income.

- **Interior (Cobb–Douglas):** tangency holds; $x^* = \frac{a}{a+b} \frac{m}{p_x}$ — constant budget shares.
- **Perfect substitutes:** compare a/p_x vs b/p_y — corner on the better-value good, or any bundle if tied.
- **Perfect complements:** no tangency; fixed proportion $x = y$, then budget $\Rightarrow x^* = m/(p_x + p_y)$.
- **Quasilinear:** x^* is set by prices alone (no income effect); spare income flows to y .

Next: let prices and income vary to trace the **demand curve**, then split a price change into substitution and income effects.